

RISK-ON SCENARIO

CLEAR SKIES & HIGH LIQUIDITY

- ROBUST GROWTH
- SURPLUS LIQUIDITY
- RISK APPETITE HIGH
- FRICTIONLESS CREDIT ACCESS

TAIL-RISK SCENARIO

LIQUIDITY SHOCK & DURATION CONFLAGRATION

- SEVERE LIQUIDITY DRAIN
- RAPID SPREAD WIDENING
- DURATION VOLATILITY SPIKE
- SYSTEMIC RISK-OFF FLIGHT

EPISTEMIC DESIGN & QUARANTINED SCAFFOLDING

This edition combines recursive learning loops with isolated hypothesis exploration. Candidate scenario variations enter as quarantined hypotheses governed by strict causal separation. Under this protocol, hypotheses inform stress-testing and challenger execution paths without mutated authority rights. Models maintain full audit traces under zero state-pollution constraints.

SOURCE BOUNDARY & EVENT ATTRIBUTION

Prints through September 26, 2026 reflect fixed scenario parameters anchoring state slice D3. The systemic extraction shock is parameterized as a continuous state-transition vector rather than an uncontrolled exterior feed. This isolation ensures replay determinism across the temporal decision harness without reliance on unvetted external telemetry.

MARKET MAYHEM

DUAL-PLANE SEPARATION INVARIANT

Observed state World/X contains immutable empirical inputs and fixed simulation injections. Belief state Belief/Z houses challenger hypotheses, model-derived valuations, and synthetic liquidity projections. Strict directional boundaries prevent uncommitted hypotheses from contaminating authoritative state.

PLANE DOMAIN	BASELINE DEFINITION (D0 ANCHOR)	CHALLENGER PATH (D3 PEAK STRESS REALIZATION)
World / X	Synthetic baseline spreads; normal funding fluidities.	Rolling extraction-shock injection; duration peak overshoot.
Belief / Z	Persistent hawkish rate repricing.	Temporary duration crest followed by sharp liquidity normalization.
Credit Channel	Controlled baseline spreads across prime and subprime tranches.	Flight-to-quality compression alongside severe synthetic friction.
Authority Rule	Authoritative reference state; frozen schema.	Quarantined simulation experiment; no direct mutation permitted.

D3 SYNTHETIC CREDIT SPREAD MATRIX

CHALLENGER STATE EVALUATED

SIMULATION METRIC TOKEN	D0 BASELINE SPREAD	D3 CHALLENGER PEAK	NET SHIFT (Δ)	PRIMARY TRANSMISSION CHANNEL
SIM-HY-CREDIT	+345 bps	+382 bps	+37 bps	Retail liquidity drain & risk-off premium
SIM-BSL	+450 bps	+468 bps	+18 bps	Institutional syndicated facility re-pricing
SIM-UNITRANCHE	+625 bps	+640 bps	+15 bps	Direct lending illiquidity buffer friction
SIM-CC0	+780 bps	+812 bps	+32 bps	Subordinated mezzanine risk-clearing hurdle

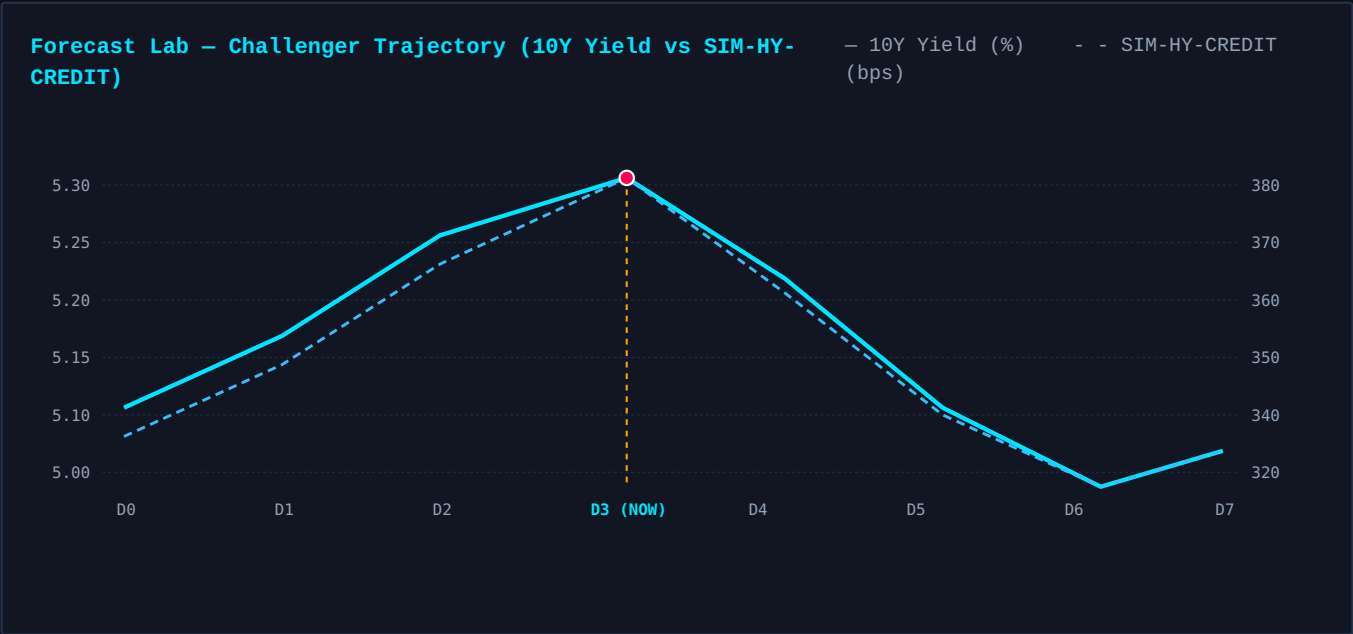
SPREAD CALIBRATION TAKEAWAY

The spread matrix acts strictly as an anchored challenger path. Infiltration of these values into baseline production tables is halted until deterministic verification passes through gate C5.

MARKET MAYHEM

Section 2: Forecast Lab & Synthetic Path Hypotheses

PROJECTION: D0 → D7 CYCLE
CURRENT PIN: D3 CREST



ID	HYPOTHESIS PREMISE & MECHANICS	FALSIFICATION CONDITION / ADVERSARIAL INVALIDATION GATE
F-01	Duration Overshoot: Challenger yields crest aggressively at D3 before rapid policy-driven reversal.	Invalidated if D4 terminal yields persist >5.25% under non-flight conditions.
F-02	Credit-Spread Divergence: Spread widening can pause even as duration volatility hits peak conflagration.	Invalidated if HY credit default swaps decouple >45 bps past unitranche ceilings.
F-03	Informativeness Metric: Spread divergence relative to liquidity decay is more predictive than raw delta.	Invalidated if multi-factor variance analysis yields attribution <0.65 to duration vector.
F-04	Convexity Dislocation: Localized geopolitical action induces a short-lived macro convexity shock.	Invalidated if cross-market equity vol stays depressed below historical 2-sigma thresholds.
F-05	Path-Dependent Normalization: Reversion gradient retains higher information entropy than peak size.	Invalidated if recovery path matches linear mean-reversion with zero lag hysteresis.

DETERMINISTIC AUTHORITY & PIPELINE CONSTRAINTS

Learning updates are strictly downstream of verification. Pipeline stages C0–C3 generate exploratory insights; gate G12 isolates hypotheses; stages C4–C5 perform deterministic arithmetic and policy validation; C6 commits an unmodifiable Decision Artifact. Learning updates are evaluated post-commit by gate G13.

EXECUTION STAGE	PRIMARY CONTROL & OPERATION	GENERATED OUTPUT & STATE TOKEN
C0–C1 Sense / Estimate	Observe X; generate ModelEvidenceEnvelope with pinned schema and cutoff.	Cryptographically signed evidence token.
G12 Isolation Gate	Partition model weights, policies, and hypotheses; block state mutation.	Isolated hypothesis sandbox.
C2–C3 Believe / Hypothesize	Synthesize belief state Z; project candidate challenger trajectories.	Candidate forecast vectors (F-01..F-05).
C4–C5 Resolve / Admit	Verify arithmetic, evaluate adversarial limits, and enforce policy boundaries.	Governed execution verdict.
C6 Commit Artifact	Generate permanent decision artifact with decision hash H_D and causal hash H_C .	Immutable Decision Artifact.
G13 Recursive Learning	Replay execution outcome; attribute delta across axes; register bounded update.	Bounded policy update candidate.

GOVERNED RECURSIVE UPDATE INVARIANT

In all learning iterations, exactly one causal dimension is varied while keeping world state, evidence records, and governance policies locked. The historical simulation is replayed, decision hashes are reconciled, and any attributable delta is measured. No probabilistic algorithm, vector score, or generative agent can mutate baseline models without deterministic sign-off.

PERTURBED AXIS	FIXED PARAMETERS	DIAGNOSTIC TEST PROCEDURE	ADMISSION SIGNAL TRIGGER
Evidence	Model, Policy, World	Temporal poisoning and evidence synthetic mutation.	Δ Decision + Full provenance audit
Model	Evidence, Policy, World	Challenger v4 run pinned against Baseline v3.	ModelDrift attribution score
Policy	Evidence, Model, World	Differential policy rule evaluation on replay.	PolicyDrift verification score
World	Evidence, Model, Policy	Counterfactual macro injection & shock parameterization.	World-shock sensitivity delta
Graph	Node values & states	Dependency disruption & topological perturbation.	Causal-path stability index
Retrieval	Downstream layers	Synthetic context corruption & embedding manipulation.	Evidence sufficiency ≥ 0.98

ADVERSARIAL EVALUATION CORPUS

PROMPT INJECTION • SYNTHETIC TEMPORAL POISONING • ENTITY COLLISION • OUT-OF-DISTRIBUTION VOLATILITY • ADVERSARIAL SPREAD MANIPULATION • POLICY GATE BYPASS • GRAPH CYCLE PERTURBATION • RETRIEVAL DRIFT • TOKEN CORRUPTION

BENCHMARK METRICS & DIAGNOSTIC MEASURES

The verification harness evaluates attribution accuracy, causal path determinism, false-alarm frequency across credit tranches, adversarial resilience, and stability against policy regressions. Any run failing reproducibility under exact seed parameters is discarded.

```
001 INIT Market Mayhem v4 harness; temporal anchor: 2026-09-26
002 RETRIEVE Qdrant context collection: credit_spreads/macro_shocks
003 VALIDATE Default probability curve (10,000 synthetic Monte Carlo iterations)
004 CROSSCHECK PennyLane & Qiskit distribution validation gates
005 INJECT Operation Absolute Resolve duration conflagration vector
006 REPLAY seed-42/D3/historical temporal state verified
007 FORECAST Synthesize hypotheses F-01..F-05 under isolation
008 VERIFY Deterministic arithmetic + policy checks completed
009 COMMIT Immutable DecisionArtifact (HD: a8f2..91c0, HC: 7e3d..44b1)
010 EVALUATE Six-axis drift decomposition across model and policy boundaries
011 LEARN G13 isolated review – bounds accepted
012 PERSIST Full provenance graph and canonical sha256 committed
```

ARTIFACT TOKEN	STRUCTURAL ROLE & DETERMINISTIC CONSTRAINT
H_D	Decision Hash: byte-stable state fingerprint; independent of thread timing.
H_C	Causal Hash: preserves step-by-step lineage isolated from evidence source data.
State Slice D3	Temporal world state at September 26, 2026; unmodifiable historical snapshot.
DecisionArtifact	Governed deliverable; cannot be overwritten by subsequent learning loops.
G13 Gate	Recursive learning guard requiring deterministic verification to update bounds.

CANONICAL ARTIFACT INTEGRITY HASH

SHA256: 82dc99a14f3b72ea91bc3501a3cf88562d90e24177cb25e89d1469e38fbb230e

PROV-0 Execution Agent: Adam05 • Telemetry trace certified under temporal epistemic control guidelines.